DSP · L02–03

Sampling, DT signals, system properties

Cheat sheet · Day 02 · Sun 25 Oct · pack D1

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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SAMPLING  f0=F0/Fs, w0=2*pi*F0/Fs. Reduce w0 into (-pi,pi]. Nyquist RATE = 2Fmax.
  Alias: Fa = F0 - k*Fs in (-Fs/2, Fs/2].  sin(-a) = -sin a (sign flips), cos(-a) = c
  Ideal DAC: substitute n = Fs*t AFTER reducing.  cos(2*pi*k*n) = 1 for integer k.
PERIODIC (DT)  f0 = k/N in lowest terms -> N.  Leftover pi -> aperiodic.  Sums: N = L
ADC  D = (xmax-xmin)/(L-1);  b = ceil(log2 L);  bit rate = b*Fs;  |eq| <= D/2;  Fmax
ENERGY/POWER  E = sum|x|^2 (finite -> energy, P=0).  P = lim 1/(2N+1) sum_{-N}^{N}|x
SEQUENCES  u(n+a)-u(n-b) = 1 on [-a, b-1].  x(n-d): shift right.  x(Mn): keep multipl
  x(n/L): zeros between samples.  x(an+b): substitute each n directly.  ALWAYS mark r
PROPERTIES  Static: only x(n).  Linear: x=0 -> y=0, no x^2/|x|/cos x/offset.
  TI: [input delayed, explicit n untouched] vs [every n -> n-d].  Explicit n coeffici
  Causal: no x(n+k), check n<0 too.  Stable: bound |y| by Mx, or u(n) -> infinity dis
  One concrete counterexample disproves a property.
CORRELATION  rxy(l) = sum x(n)y(n-l) (no flip).  rxy(l) = ryx(-l).  rxx even, rxx(0)
  Radar: d_hat = argmax |ryx(l)|;  d = round(2*Fs*r/c).
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